Chapter 27 | Written solutions

Time-Series Forecasting

All eleven exercises are solved here. We begin with what is known when a forecast is issued, build windows and patches by hand, and then check forecasts and evaluation protocols in Python.

Three plots make the main distinctions visible: asymmetric quantile loss, seasonal drift, and interval coverage versus width. All numerical results use small teaching fixtures, not performance measurements of TFT, PatchTST or a pretrained model.

1. Separate what is known from what will arrive

The forecast is issued at Thursday's close. A future date does not automatically make an input unavailable: a published opening schedule can already be known. Conversely, a historical timestamp does not prove that a value has arrived.

Value	Input type	Available at Thursday's close?
Store size	Static covariate	Yes, if already recorded
Friday's scheduled opening hours	Known future covariate	Yes, use the schedule published by the origin
Thursday's sales	Past observation	Yes, assuming the daily total has arrived
Friday's measured foot traffic	Future observation	No

The first three may enter the forecast under these assumptions. Preserve the schedule version that existed then. If Thursday's total is finalized later, use the latest available total or a predefined missing-value rule. The code illustrates the availability test with synthetic arrival times; it does not infer availability from the target date.

2. Keep targets inside their split

The last training target must be at most 15. With three targets, the last training origin is therefore 13: its targets are 13, 14 and 15. Origins 14 and 15 cross the split and must be omitted. The first test origin is 16, whose history can legitimately include training-period observations.

Window	Origin	Past indices	Target indices
Last training	13	9, 10, 11, 12	13, 14, 15
First test	16	12, 13, 14, 15	16, 17, 18

For split boundary $b = 16$, training origins satisfy $o + H \leq b$ and test origins satisfy $o \geq b$. The final complete test origin is 21, with targets 21–23. Sharing past observations across windows is allowed; using a future target as an earlier input is not.

3. Trace overlapping patches

The starts are 0, 2 and 4. Each patch contains three consecutive values.

Patch	Indices	Values
First	0, 1, 2	4, 2, 6
Second	2, 3, 4	6, 1, 7
Third	4, 5, 6	7, 3, 8

The first two patches share index 2, value 6. The last two share index 4, value 7. Index 7, value 5, appears in no complete patch: a fourth patch would start at 6 and require index 8.

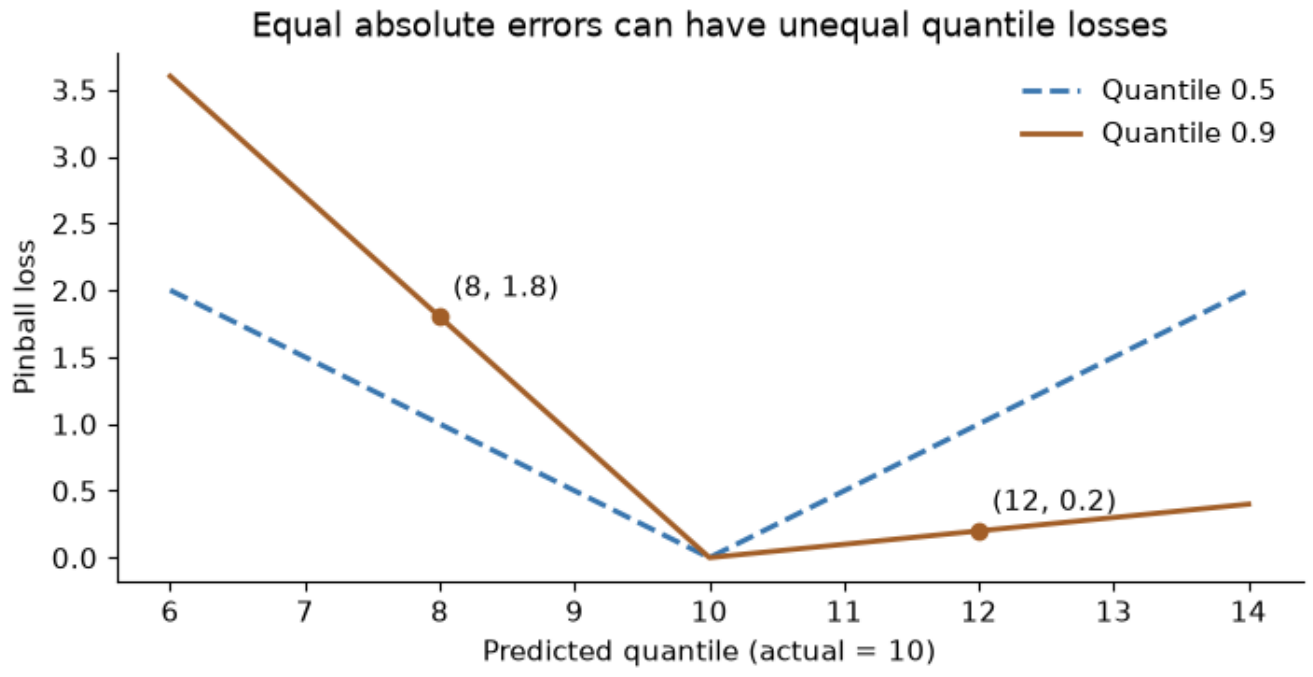
The formula gives $1 + \lfloor (8 - 3)/2 \rfloor = 3$ patches. Dense attention has $3^2 = 9$ ordered query–key pairs per channel and head, including three self-pairs. Both directions between two different patches count. This is a pair count, not a measurement of total runtime. Padding or changing the stride would change the tokenization and must be stated.

4. Count interval hits

The actual value 6 lies on the upper endpoint of $[2, 6]$ and counts as a hit. The value 8 lies above $[4, 7]$ and misses. Coverage is $1/2 = 50\%$. Widths are 4 and 3, so mean width is $(4 + 3)/2 = 3.5$ target units.

For the quantile calculation, use residual $e = y - \widehat{q}$. With actual 10, forecast 8 gives $e = 2$ and loss $0.9(2) = 1.8$. Forecast 12 gives $e = -2$ and loss $(1 - 0.9)(2) = 0.2$. Both absolute errors are 2; underprediction costs nine times as much for this high quantile.

The plot shows both sides of that penalty and compares it with the symmetric 0.5-quantile loss. A quantile loss value is not an interval coverage rate. A nominal 80% interval also does not promise that a two-observation sample will have exactly 80% hits.



5. Inspect rolling windows

Imagine pausing the series just before the observation at index 3. The three values already available are $(0, 1, 2)$. The next two values, $(3, 4)$, are the targets we will score against later. The **forecast origin** is the index of the first unknown value, so this window has origin 3. The past ends at index 2.

For an origin t, a lookback of L uses indices $t - L$ through $t - 1$. A horizon of H uses targets from t through $t + H - 1$. In a series of length N with indices 0 through $N - 1$, the first past index must satisfy $t - L \geq 0$, and the last target must satisfy $t + H - 1 \leq N - 1$. Together these give $L \leq t \leq N - H$. Here $L = 3$, $H = 2$, and $N = 10$, so the valid origins are 3 through 8.

At origin 8 the past is $(5, 6, 7)$ and the targets are $(8, 9)$. The last target is $8 + 2 - 1 = 9$, the final index in the input. Origin 9 would need targets at indices 9 and 10, so it would have only one observed target. The rows here show values, which happen to equal their indices because the series is 0 through 9.

To check every row independently, slide a block of five values through the input. Split each block after its first three values. For example, the block $(1, 2, 3, 4, 5)$ gives past $(1, 2, 3)$ and targets $(4, 5)$ at origin 4.

6. Forecast beyond a season

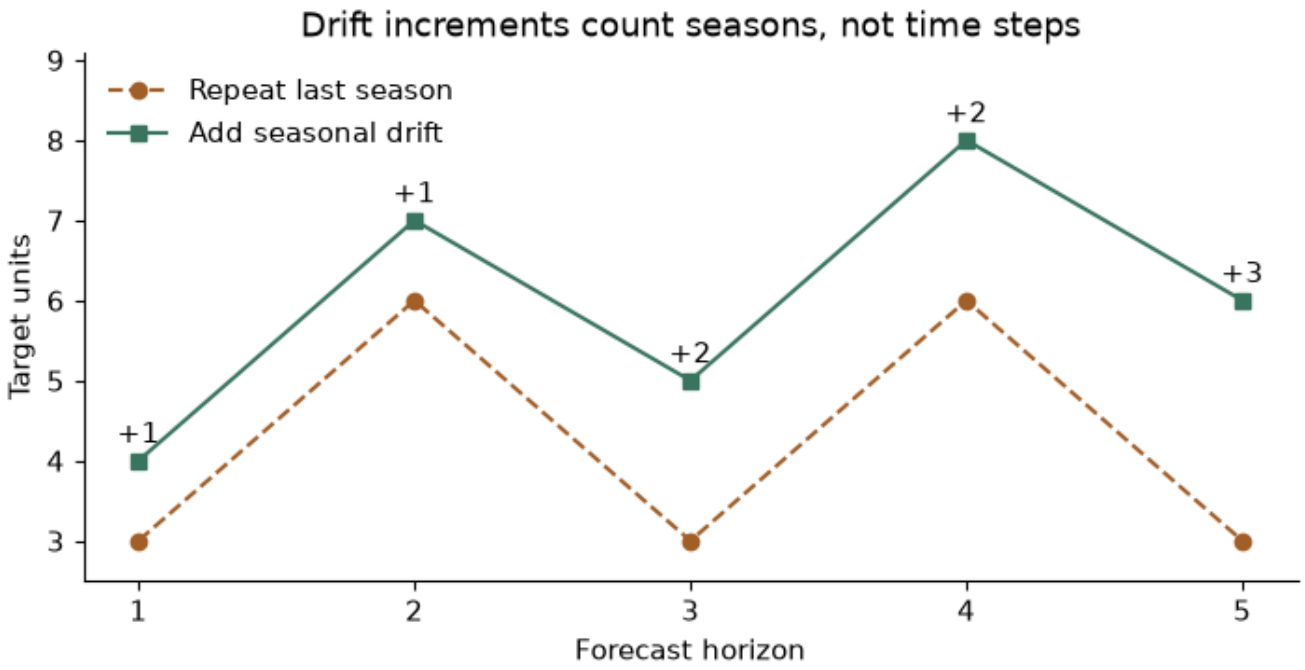
The history $(2, 5, 3, 6)$ contains two seasons of two values each. A season is one full repetition of the pattern. The seasonal-naive forecast copies the latest season, $(3, 6)$. It repeats that pair until it has five predictions, giving $(3, 6, 3, 6, 3)$.

Drift is the estimated change from one season to the next. To find it, compare each position in the second season with the same position in the first. The differences are $3-2 = 1$ and $6-5 = 1$, so their mean is $d = (1+1)/2 = 1$. This is one unit per **season**, which lasts two steps here.

The first two forecasts are one season ahead of their matching observations and receive d each. Horizons 3 and 4 are two seasons ahead and receive $2d$ each. Horizon 5 receives $3d$.

Horizon 3 is the first prediction to receive two drift increments. The first forecast cycle gives $(3+1, 6+1) = (4, 7)$. The third prediction returns to the first position of the season, giving $3 + 2(1) = 5$. The fourth gives $6 + 2(1) = 8$, and the fifth starts the third cycle with $3 + 3(1) = 6$.

For a separate check of the drift forecast, extend the history one step at a time. For each new value, take the value two steps back and add one unit. The first two new values use the observed pair $(3, 6)$. For horizon 3, the value two steps back is the first prediction, 4, so the new value is $4 + 1 = 5$. This step-by-step calculation checks all five predictions.



7. Change what the baseline learned

At origin 24, the shop has observed values through index 23. The chapter's seasonal-naive forecast copies the last observed values for the three matching seasonal positions. It predicts $(20, 24, 21)$, while the actual values are $(24, 28, 25)$. Each prediction misses by four units. A model that learned a drift of four would add four to each prediction.

Run the chapter's example as a module, as in the listing. The notebook uses the current Python interpreter and reads the JSON from a temporary directory. The mean absolute error (MAE) in each column averages the absolute errors at that horizon across all six test origins.

The fixture adds a repeating four-position pattern to the time index. Advancing four steps returns to the same pattern value and raises the series by four. This explains why the fitted drift is four and why it removes all the errors in this fixture. That exact correction depends on how this toy series was made.

Now replace the fitted drift with zero. Origin o is the first unknown index, so `values[:origin]` ends at $o - 1$. At later origins, earlier test observations have arrived and can enter the history. The three values at indices o, $o + 1$, and $o + 2$ are still withheld for scoring.

With period four, the first three steps copy indices $o - 4$, $o - 3$, and $o - 2$. They are all in the first forecast season, so the drift model adds one increment to each. With a zero increment, those three predictions stay equal to the seasonal-naive predictions. Check both forecasts at every origin against the values at the three source indices.

All six forecasts with zero drift match seasonal naive entry for entry. Both methods have MAE four at every horizon. Only the fitted version adds the change the series makes between seasons. The extra history at later origins updates the values the model copies. The drift used in the original run stays fixed at the value learned from the training prefix.

8. Compare interval widths

Imagine issuing the same point forecast with three different margins. The point forecast can be equally accurate in every case, while the wider interval catches an actual value that fell outside the narrower one.

Use the chapter's two-origin fixture. The actual rows are $(10, 12, 15)$ and $(11, 13, 14)$, and the prediction rows are $(10, 11, 14)$ and $(10, 13, 12)$. The first column contains two one-step forecasts, the second two two-step forecasts, and the third two three-step forecasts. Comparing actual values to point predictions gives absolute errors

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Averaging down the columns gives MAE $(0.5, 0.5, 1.5)$. For half-width w, subtract w from each prediction to get the lower endpoint and add w to get the upper one. Each interval then has width $2w$. Coverage at a horizon is the fraction of its two actual values in their respective intervals, including both endpoints. An actual value is covered exactly when its absolute error is at most w.

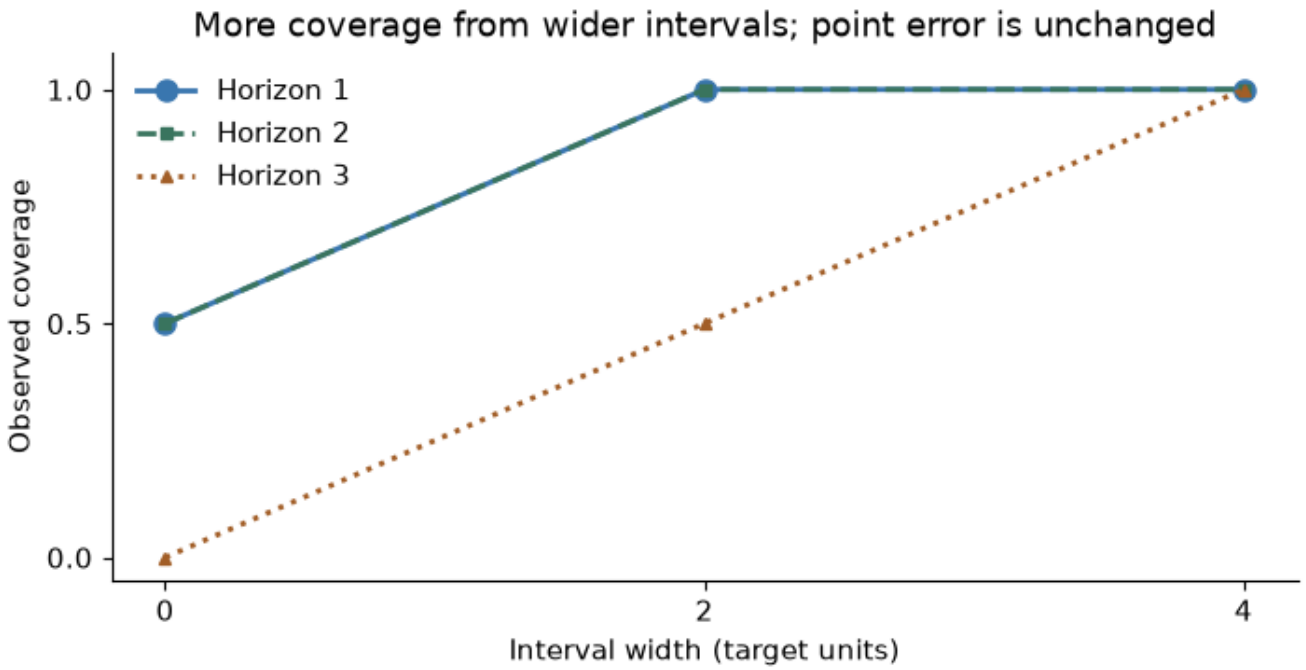
Half-width	Horizon 1 coverage	Horizon 2 coverage	Horizon 3 coverage	Mean width at every horizon
0	$1/2$	$1/2$	$0/2$	0
1	$2/2$	$2/2$	$1/2$	2
2	$2/2$	$2/2$	$2/2$	4

The intervals with half-width zero contain only exact predictions. At half-width one, errors of one are covered because the endpoints count. At half-width two, even the largest error is covered. MAE depends only on the actual values and point predictions, so we expect it to stay fixed.

For the second origin at horizon three, the point prediction is 12 and the actual value is 14. At half-width one, the upper endpoint is 13 and the interval misses. At half-width two, the upper endpoint is 14 and the interval hits. That change brings coverage at horizon three from one hit out of two to two hits out of two. The error for this same point prediction stays at two units. The wider interval buys more coverage at the cost of precision about where the outcome will fall.

Widening these intervals after inspecting the same outcomes is a sensitivity exercise. It cannot establish prospective 80% coverage. A real protocol would choose the interval rule using training or earlier calibration outcomes, issue new intervals, and score them only when their outcomes arrive. With two observations per horizon, possible observed coverages are only 0%, 50% and 100%.

The plot uses exactly the three rows of the table. MAE stays at (0.5, 0.5, 1.5) for every width; only the intervals change.



9. Audit feature availability

At origin o, suppose sales through $o-1$ have arrived. A centered seven-day average stored on row $o-1$ uses days $o-4$ through $o+2$. The last three days are precisely the withheld targets of a three-step forecast. A historical row label therefore hides future information.

Use a trailing average of days $o-7$ through $o-1$, ending earlier if records arrive with delay. Define a missing-history policy in advance. This origin-time summary can be reused for all horizons; computing a fresh trailing average at each future date would eventually require unknown sales.

For weather, preserve both issue time and arrival time, location and target date for each forecast version. At origin o, select the latest forecast for the requested date that had actually arrived by o. Keep actual measured weather in a separate field. An update issued on $o+1$ is too late for a forecast originally made at o, even if it predicts $o+2$. If historical forecasts are unavailable, do not substitute the later measurements; use a predefined missing-value rule or omit that feature.

The code perturbs all observations at and after one origin. The trailing feature stays unchanged, while the centered feature changes. This demonstrates leakage in this construction; passing one perturbation test alone would not audit a whole pipeline.

10. Choose a rolling protocol

Consider a daily series and a three-step horizon. Use days 0–167 for training, 168–195 for validation and 196–251 for test. A window belongs to a split only when all its targets lie inside it. During validation, replay the fixed and weekly-refit procedures in time order. Choose lookback, features and update settings there, then freeze those choices.

Before origin 196, fit both final systems on the same permitted training-plus-validation pool, with targets fully observed by then. This refit after model selection is part of the declared protocol. Test origins are 196–249: 54 forecasts per system. The last origin's targets are 249, 250 and 251. Give both models the same observed history, future calendar, and as-issued plans or weather forecasts; exclude future measured traffic and sales.

The fixed system retains its parameters and fitted transformations. The weekly system refits at origins 203, 210, 217, 224, 231, 238 and 245, using only targets that have arrived by each update. This gives seven refits after the initial fit. Recompute training-fitted transformations within the allowed pool at each refit. Both models may refresh their input history daily; this does not itself update their parameters. Earlier test outcomes may train later weekly refits under this protocol, but no issued forecast may be revised.

Compare paired errors at the same origins separately by horizon. The 162 forecast–target pairs contain only 56 distinct target days, so do not treat all pairs as independent replicates. Use time-aware uncertainty estimates. Report initial fit time, all seven update times, total update cost, prediction latency and hardware. State whether an update finishes before that day's forecast deadline; a late model update cannot support an on-time prediction retroactively.

11. Evaluate intervals under a shift

Let d be the opening date and use a seven-day forecast window, comparing horizons one and seven on the same origins. A before-origin satisfies $o + 6 < d$; an after-origin satisfies $o \geq d$. Origins $d - 6$ through $d - 1$ straddle the opening and form a separate transition group. At $o = d - 2$, horizon one targets an old-regime day but horizon seven targets $d + 4$. Pooling that row into either main group would mix regimes.

Fix study lengths and weekday balance before examining scores. Fit the point model before the study. Use its forecasts on a distinct earlier calibration period to estimate the 10th and 90th percentiles of signed residuals $y - \widehat{y}$, separately by horizon. Every calibration outcome must arrive before the first evaluated forecast. Add those frozen offsets to future point predictions. This is one nominal 80% interval construction; estimated residual quantiles do not guarantee 80% coverage after a distribution shift.

Keep the model fitting and input-history policy fixed across the comparison. Save each issued point forecast and interval with its timestamp. For each horizon and regime, report hits, total forecasts, coverage and mean width. Retain transition rows as a separate result. With fixed additive offsets, width stays constant within a horizon: a coverage drop then means more misses at that same width. If adaptive calibration is studied, declare it as another procedure and update only from outcomes already available.

The synthetic calculation uses calibration residuals from -4 through 4 at horizon one and twice those values at horizon seven. NumPy's stated linear quantile convention yields offsets $(-3.2, 3.2)$ and $(-6.4, 6.4)$. Before-shift residuals cycle through -4 to 5 twice, giving 14/20 hits at each horizon. Shifting them by 4 at horizon one and 8 at horizon seven leaves 8/20 hits. Widths remain 6.4 and 12.8. These small constructed counts illustrate the scoring rules; they are not evidence about a delivery service.

Coverage here is marginal at each horizon. It does not measure whether every day in a seven-day path is covered at once. For example, independent daily 80% hits would give $0.8^7 \approx 21\%$ whole-path coverage; dependence changes that calculation.

Checks and next steps

All eleven solutions connect an independent calculation to the chapter's functions or an explicitly specified evaluation design. The saved plots and outputs separate timing, point accuracy, interval coverage and update cost.

Try delaying every observation by one day. Redefine the available-history boundary before running the forecaster, and verify that changing unseen future values cannot change its issued prediction. This tests the availability contract, which matters before choosing a more complex architecture.